AI & Software Engineering Workshop

Week 2, Day 3: Training, Part 1

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Press the button

```
python 4_train.py --preset xlarge --tokenizer bpe
```

- **350M parameters · 36K steps · ~8 hours on this GPU**
- It will *not* finish during class; that's by design
- First milestone: watch the loss start falling

We start it **now**, then learn what it's doing while it does it.

One training step, slow motion

1. **Batch**: grab a chunk of tokenized text
2. **Forward pass**: model predicts every next token
3. **Loss**: how wrong were the predictions? One number.
4. **Backward pass**: trace every weight's share of the blame
5. **Update**: nudge all 350M weights a tiny step downhill

Repeat 36,000 times. That's the whole secret.

Words you'll see scroll by

- **Loss**: wrongness. Down = learning.
- **Learning rate**: step size downhill. Too big: chaos. Too small: forever.
- **Perplexity**: "how many tokens was the model torn between?" Lower is better.
- **Checkpoint**: full snapshot of the weights, saved **every 1000 steps**

Watch a model grow up

Generate from checkpoints as they appear:

- step 1000: character soup
- step 5000: Armenian-shaped words
- step 20000: grammar emerges
- step 36000: fluent continuation *(tomorrow)*

Keep your favorite "baby pictures" to compare on Friday.

Pause and resume: practice now

```
# Ctrl+C  →  saves a checkpoint on the way out

python 4_train.py --preset xlarge --tokenizer bpe \
    --resume_from checkpoints/step_5000.pt
```

- Checkpoint = weights + optimizer + step counter + schedule
- Resume continues **exactly** where it stopped: nothing lost
- This is your insurance against power cuts, crashes, and impatience

Tonight

- Machines **stay on, training**: do not shut down 🌙
- ~8 hours of GPU time happens while you sleep
- Tomorrow morning: a model that's been studying all night

Today → Tomorrow

Today: training started, and you can read its vital signs.

Tomorrow: what's actually inside (attention, layers, loss curves) and your model's first real report card.